



Antenna Matching for UHF – RFID Transponder ICs

Overview

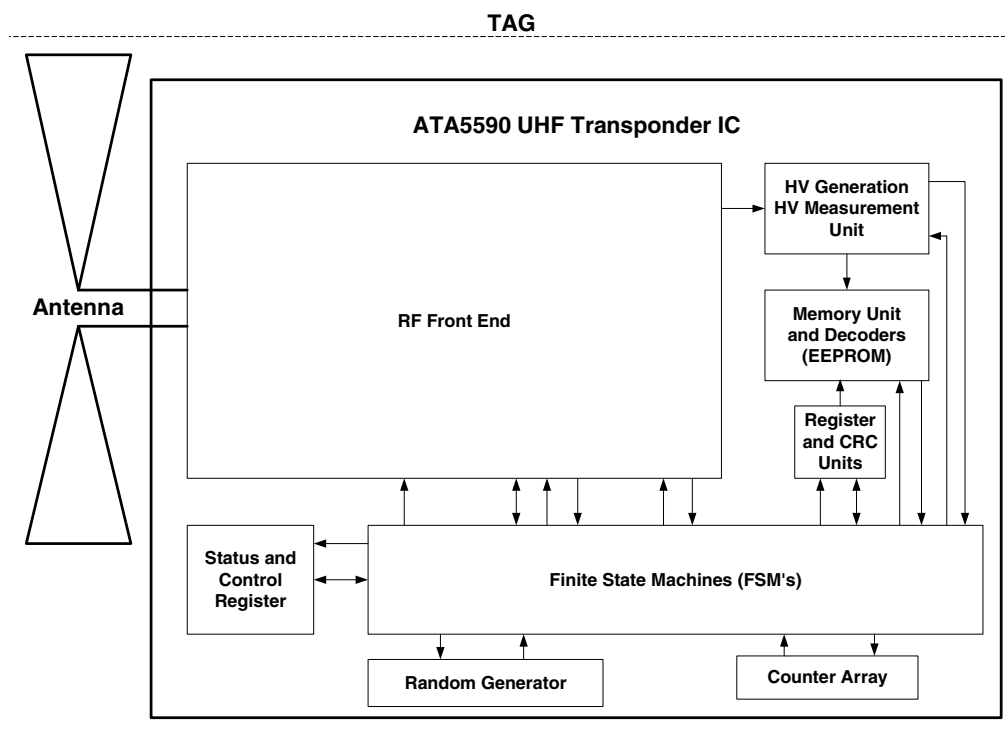
To achieve high performance with regards to the read and write distance of a tag, it is essential that the tag–antenna design be matched to the ATA5590 input impedance, because the return link modulation of the ATA5590 uses PSK (Phase Shift Keying). The main advantages of PSK are related to

- higher SNR (Signal to Noise Ratio)
- lower BER (Bit Error Rate)

and result in increased communication distance.

In PSK the information is in the delta of the phase angle, $\Delta\phi$; therefore, it is essential to properly match the antenna to get a maximum $\Delta\phi$ that can easily be detected by any I/Q demodulator. The return link is always one of the most critical criteria in UHF-RFID and limits the operation range of many application systems.

Figure 0-1. Block Diagram of the UHF Transponder



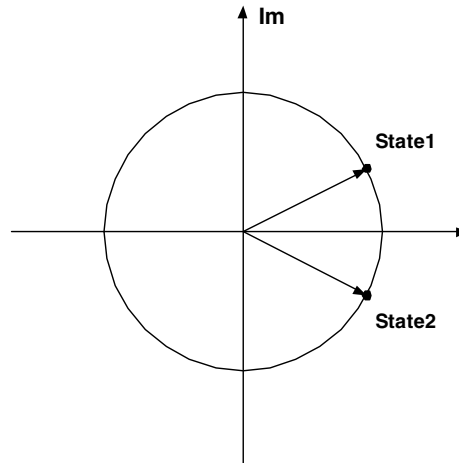
Rev. 4843A–RFID–02/05



1. PSK Modulation and the Resulting Change of the Phase Angle, $\Delta\phi$

One possibility of modulating a digital signal is to shift the phase depending on the data stream. Digital phase modulation, or PSK, is a collective term for a huge class of modulation methods. The simplest way of modelling a signal with PSK is to make the phase change dependent on only two discrete conditions. With this method, the phasing of the data signal carrier is alternated between two phase states; this technique is identified as BPSK (Binary Phase Shift Keying).

Figure 1-1. BPSK Mapping and the Corresponding Logical Values



	Modulator
State 1:	state 1
State 2:	state 2

If the real part of the IC impedance is nearly the same for both states ($Re_{t1} = Re_{t2}$), the value of the phase angle $\Delta\phi$ depends only on the change of the imaginary part of the IC (see Figure 1-1). To achieve the highest $\Delta\phi$ it is essential to address low real resistance of the tag antenna impedance. Therefore it is important to achieve a low real-part impedance over the target frequency range. Reducing the real-part impedance increases the Q-value, and therefore reduces the bandwidth of the transponder. It will still result in a compromise.

In example 1, the resonance frequency is in the middle of the two states of the tag – this results in a maximum phase angle. Then the backscattered power is equal for both states.

1.1 Formula

IC impedance state 1: $Z_{IC1} = R_{et1} - jIm_{t1}$

IC impedance state 2: $Z_{IC2} = R_{et2} - jIm_{t2}$

Antenna impedance: $Z_{ant} = R_a + jIm_a$

Resulting impedance (IC + antenna) when optimal power matched: $Z_{total} = 2 \times R_e$

Phase angle $\Delta\phi$: $\Delta\phi = |\phi_1 - \phi_2|$

$$\tan\phi_1 = \frac{\text{Im}_{t1} + \text{Im}_a}{\text{Re}_{t1} + \text{Re}_a}$$

$$\tan\phi_2 = \frac{\text{Im}_{t2} + \text{Im}_a}{\text{Re}_{t2} + \text{Re}_a}$$

1.1.1 Example 1

State 1: $Z_{IC1} = 6 - j202$

State 2: $Z_{IC2} = 7 - j216$

One approach would be to match the antenna impedance into the middle of both states:

$Z_{ant} = 6 + j209$

Modulation on (state 1), results in an IC impedance of: $Z_{total} = 12 + j7 \rightarrow$ pointer 1

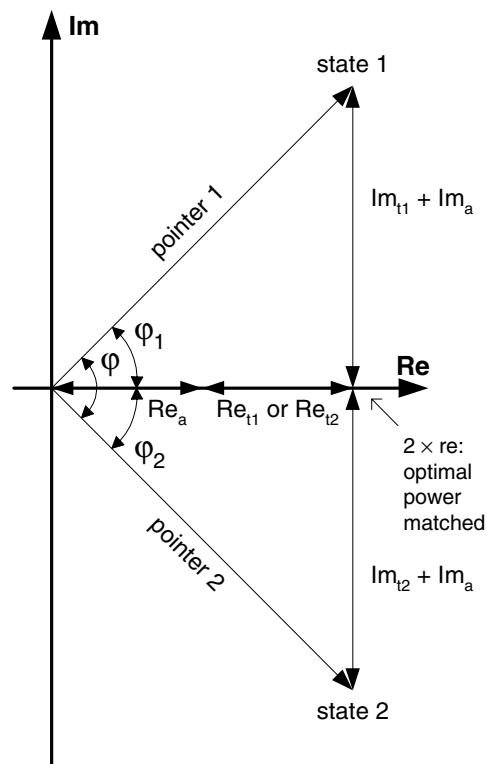
Modulation off (state 2), results in an IC impedance of: $Z_{total} = 13 - j7 \rightarrow$ pointer 2

$\tan(\phi_1) \approx -0.58 \rightarrow \phi_1 \approx -30^\circ$

$\tan(\phi_2) \approx 0.54 \rightarrow \phi_2 \approx 28^\circ$

$\Delta\phi = |\phi_1 - \phi_2| \approx 58^\circ$

Figure 1-2. Matching 1



1.1.2 Example 2

State 1: $Z_{IC1} = 6 - j202$

State 2: $Z_{IC2} = 7 - j216$

Another approach would be to match to state 1: $Z_{ant} = 6 + j202$

Modulation on (state 1), results in an IC impedance of: $Z_{total} = 12 \rightarrow$ pointer 1

Modulation off (state 2), results in an IC impedance of: $Z_{total} = 13 - j14 \rightarrow$ pointer 2

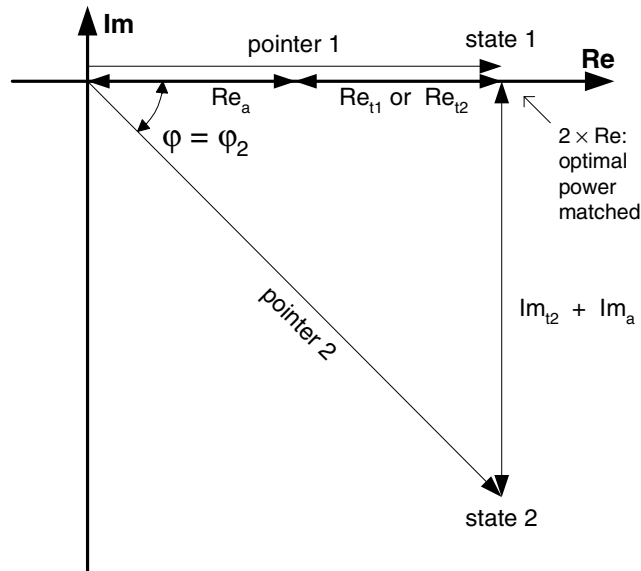
$\tan(\varphi_1) \approx 0 \rightarrow \varphi_1 \approx 0^\circ$

$\tan(\varphi_2) \approx -1.07 \rightarrow \varphi_2 \approx -47^\circ$

$\Delta\varphi = |\varphi_1 - \varphi_2| \approx 47^\circ$

Conclusion: The resulting $\Delta\varphi$ is less than with the first approach.

Figure 1-3. Matching 2



2. Input Impedance as a Function of the Carrier Frequency

2.1 Input Impedance from Die

Table 2-1. Primary Frequency Dependent Parameters at 25°C and 868 MHz

	Conditions/Comment	Typical		Unit
		Serial Impedance	Equivalent Parallel Impedance	
Input impedance ⁽¹⁾	11 μ W RF power, modulation on (state 2), idle mode	7 – j216	6970 – j216	Ω
	Modulation off (state 1), idle mode	6 – j202	6695 – j202	Ω
Input impedance ⁽¹⁾	25 μ W RF power, programming	12 – j217	3840 – j217	Ω

Note: 1. Impedance is calculated as a series impedance of R and C.

Table 2-2. Primary Frequency Dependent Parameters at 25°C and 915 MHz

	Conditions/Comment	Typical		Unit
		Serial Impedance	Equivalent Parallel Impedance	
Input impedance ⁽¹⁾	12 μ W RF power, modulation on (state 2), idle mode	7 – j205	6097 – j205	Ω
	Modulation off (state 1), idle mode	6 – j191	5706 – j191	Ω
Input impedance ⁽¹⁾	27 μ W RF power, programming	10 – j191	3552 – j191	Ω

Note: 1. Impedance is calculated as a series impedance of R and C.

Table 2-3. Primary Frequency Dependent Parameters at 25°C and 960 MHz

	Conditions/Comment	Typical		Unit
		Serial Impedance	Equivalent Parallel Impedance	
Input impedance ⁽¹⁾	13 μ W RF power, modulation on (state 2), idle mode	7 – j196	5827 – j196	Ω
	Modulation off (state 1), idle mode	6 – j182	5436 – j182	Ω
Input impedance ⁽¹⁾	27 μ W RF power, programming	10 – j182	3424 – j182	Ω

Note: 1. Impedance is calculated as a series impedance of R and C.

The IC is optimized for operation at carrier frequencies in the UHF band, but it is also possible to use the same circuit for microwave-based systems (2450 MHz).

Table 2-4. Primary Frequency Dependent Parameters at 25°C and 2450 MHz

	Conditions/Comment	Typical		Unit
		Serial Impedance	Equivalent Parallel Impedance	
Input impedance ⁽¹⁾	48 μ W RF power, modulation on (state 2), idle mode	4 – j85	1810 – j85	Ω
	Modulation off (state 1), idle mode	4 – j79	1564 – j79	Ω
Input impedance ⁽¹⁾	85 μ W RF power, programming	5 – j85	1450 – j85	Ω

Note: 1. Impedance is calculated as a series impedance of R and C.

2.2 Input Impedance from TSSOP10 Package

Table 2-5. Primary Frequency Dependent Parameters at 25°C and 868 MHz

	Conditions/Comment	Typical		Unit
		Serial Impedance	Equivalent Parallel Impedance	
Input impedance ⁽¹⁾	Modulation on (state 2), idle mode	12 – j159	2119 – j159	Ω
	Modulation off (state 1), idle mode	11 – j145	1922 – j146	Ω

Note: 1. For packaged parts, the imaginary part decreases because of the influence of housing and bonding.

Table 2-6. Primary Frequency Dependent Parameters at 25°C and 915 MHz

	Conditions/Comment	Typical		Unit
		Serial Impedance	Equivalent Parallel Impedance	
Input impedance ⁽¹⁾	Modulation on (state 2), idle mode	11 – j150	2056 – j151	Ω
	Modulation off (state 1), idle mode	10 – j136	1860 – j137	Ω

Note: 1. For packaged parts, the imaginary part decreases because of the influence of housing and bonding.

Table 2-7. Primary Frequency Dependent Parameters at 25°C and 960 MHz

	Conditions/Comment	Typical		Unit
		Serial Impedance	Equivalent Parallel Impedance	
Input impedance ⁽¹⁾	Modulation on (state 2), idle mode	11 – j142	1844 – j143	Ω
	Modulation off (state 1), idle mode	10 – j128	1648 – j129	Ω

Note: 1. For packaged parts, the imaginary part decreases because of the influence of housing and bonding.

ATA5590 Tag Antenna Matching

Table 2-8. Primary Frequency Dependent Parameters at 25°C and 2450 MHz

	Conditions/Comment	Typical		Unit
		Serial Impedance	Equivalent Parallel Impedance	
Input impedance ⁽¹⁾	Modulation on (state 2), idle mode	10 – j34	126 – j37	Ω
	Modulation off (state 1), idle mode	10 – j28	126 – j31	Ω

Note: 1. For packaged parts, the imaginary part decreases because of the influence of housing and bonding.

Note: The IC impedance depends on the used baud rate in an application system. A higher baud rate increases the current consumption, and the input impedance behavior also increases proportional to the current consumption of the IC. It is indispensable to know the baud rate in an application system to properly match the tag-antenna.

Also, all write commands need more current consumption than read commands – this also decreases the communication distance.

3. Diagrams for Tag-antenna/IC Matching

Figure 3-1 for dies and Figure 3-2 on page 8 for TSSOP10, display an acceptable high and low imaginary impedance characteristic of the tag and optimal imaginary part of the antenna. The right y-axis represents the real part of the antenna. Imaginary part high and low (state 1 and state 2) indicate the two modulation conditions of the tag (whole system). The optimum imaginary part of the antenna is matched into the middle of both imaginary states of the IC. Note that the tag-antenna also has a tolerance that influences the antenna matching.

Figure 3-1. Dies: High and Low Imaginary Part of the Tag and Optimal Imaginary Part of the Antenna; the Y-axis on the Right Side Relates to the Real Part of the Antenna

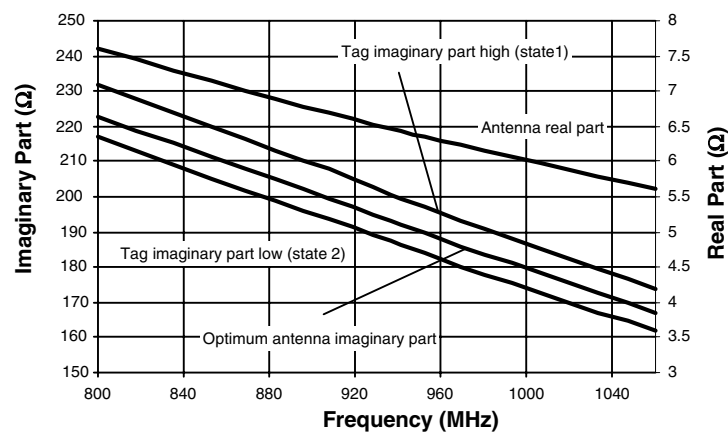
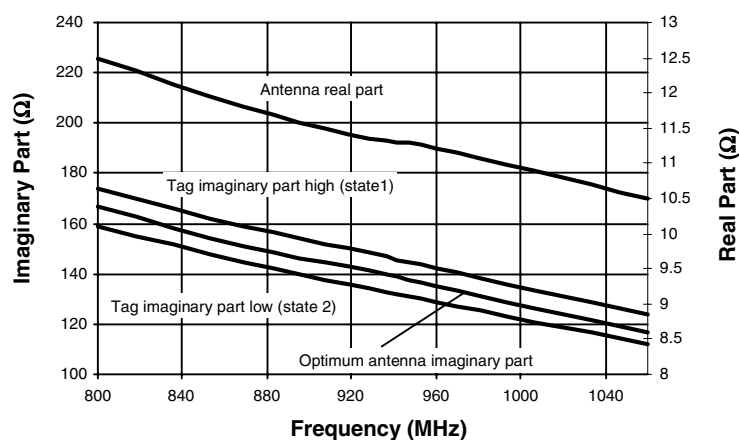


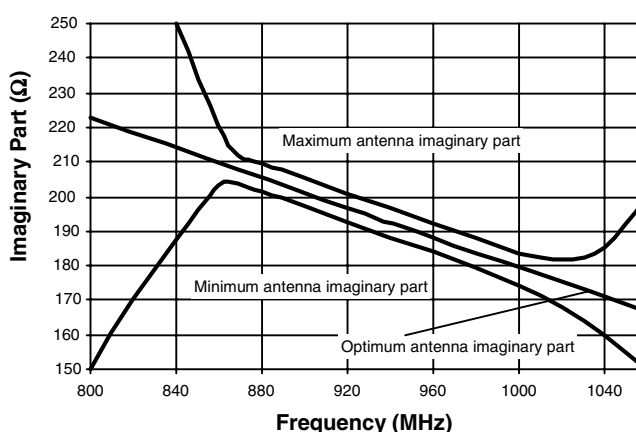
Figure 3-2. TSSOP10: High and Low Imaginary Part of the Tag and Optimal Imaginary Part of the Antenna; the Y-axis on the Right Side Relates to the Real Part of the Antenna



Note: 4 Ω deviation from the optimum imaginary part of the antenna reduces the operation range by approximately 10%, 8 Ω deviation by 20%, and 12 Ω deviation by 26%!
The range of the real part should be 10 Ω to 15 Ω where no excessive loss of operation range occurs, but a higher real part reduces the delta of the phase angle, $\Delta\phi$, of the backscatter signal

Figure 3-3 and Figure 3-4 on page 9 show an optimum antenna imaginary part in conjunction with the minimum and maximum imaginary part of the antenna. This optimum antenna imaginary part is only theoretical and can't be transferred in real life. To improve the performance of the tag, it is necessary that the antenna's imaginary part approximates the optimum antenna imaginary part, see Figure 3-3 and Figure 3-4 on page 9. A suggested approximation is shown in Figure 3-5 on page 9 for bare chips and Figure 3-6 on page 9 for TSSOP10 packages.

Figure 3-3. Die: Optimum Imaginary Antenna Part



ATA5590 Tag Antenna Matching

Figure 3-4. TSSOP10: Optimum Imaginary Antenna Part

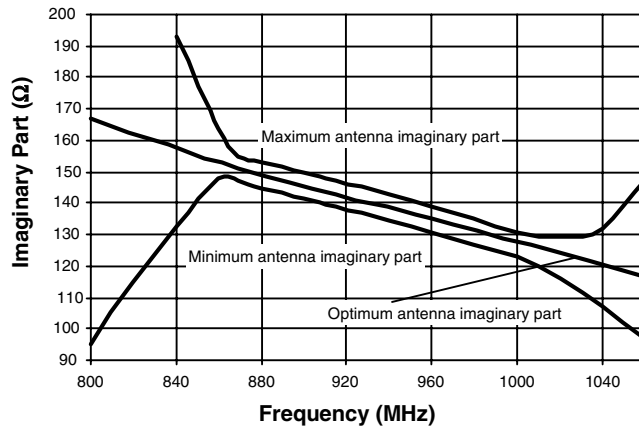


Figure 3-5. Die: Possible Antenna Imaginary Part – Good Matching for 868 MHz, Reduced Operation Range at 915 MHz and 960 MHz

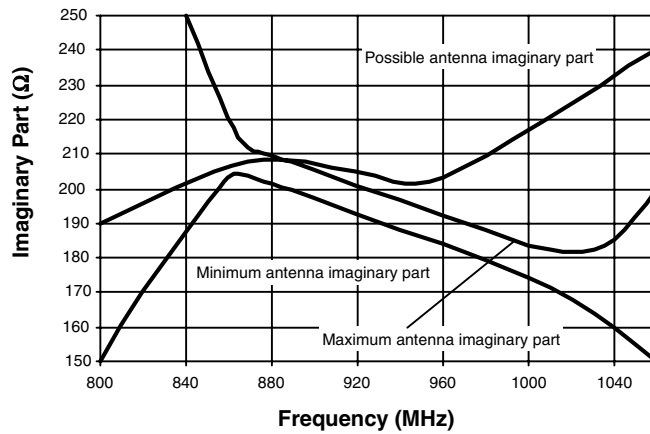


Figure 3-6. TSSOP10: Possible Antenna Imaginary Part – Good Matching for 868 MHz, Reduced Operation Range at 915 MHz and 960 MHz

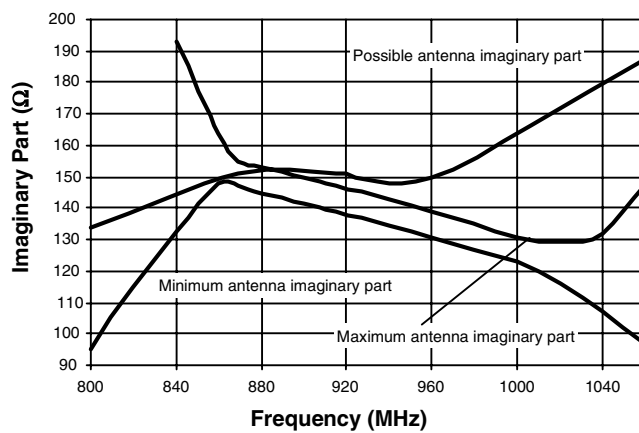


Figure 3-5 and Figure 3-6 on page 9 represent nearly optimum antenna matching for a carrier frequency of 868 MHz. There is reduced operation range at 915 MHz and at 960 MHz: the operation range decreases approximately to 30%, due to non-optimum matching. Moreover, through the assembly (for example, the housing of the die, or the connection between IC and tag antenna) additional parasitic capacitance (typically 50 fF to 200 fF), which is parallel, increases the tag-antenna capacitance and decreases the tag imaginary part. An additional capacitance between ANT and GND will also be critical, because the assembly and the backdrop of the antenna will also have a strong influence on the parasitic capacitance.

Figure 3-7. Maximal Possible Phase Angle $\Delta\phi$

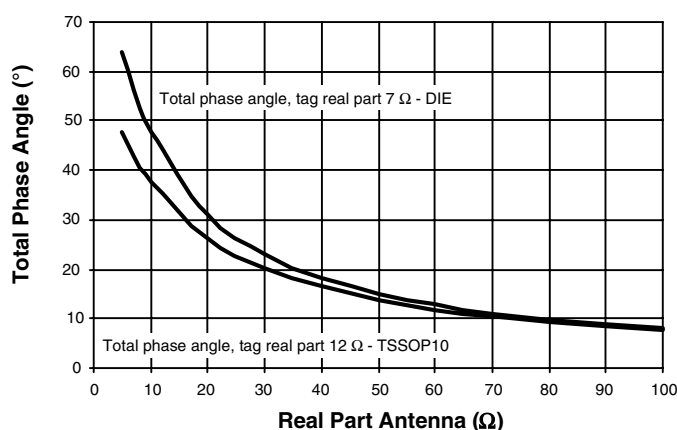


Figure 3-7 displays the maximal possible phase angle $\Delta\phi$ (for die and TSSOP10 package) depending on the following conditions for the die:

- Δ imaginary part of the bare chip: $\frac{\Delta Im}{2} \approx j7 \Omega$ – stays constant
- real part of the bare chip: 7Ω – stays constant

For the TSSOP10 package:

- Δ imaginary part of the package: $\frac{\Delta Im}{2} \approx j7 \Omega$ – stays constant
- real part of the package: 12Ω – stays constant
- real part of the antenna (from 5Ω to 100Ω)

As can be seen in Figure 3-8 and Figure 3-9 on page 11, $\Delta\phi$ decreases when increasing the real part of the antenna.

Figure 3-8. Phase Angle $\Delta\phi$ versus Imaginary Part of the Antenna – Consideration only Valid for Dies

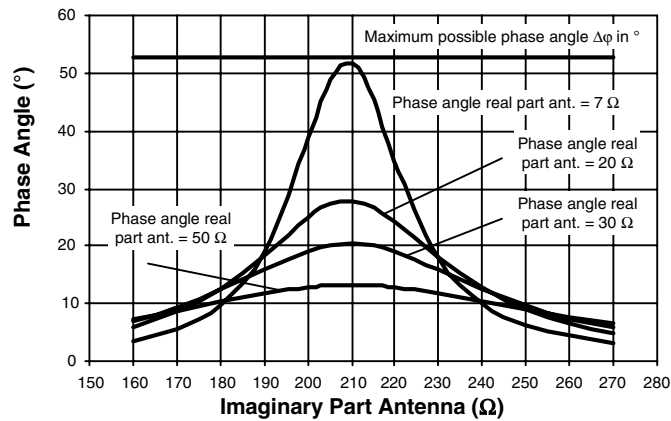


Figure 3-8 shows the phase angle $\Delta\phi$ (for die) versus the imaginary part of the antenna with the following conditions:

- Real part of the IC (state 1 and 2): $\approx 7 \Omega$ – stays constant
- Imaginary part of the IC (state 1): $\approx 202 \Omega$ – stays constant
- Imaginary part of the IC (state 2): $\approx 216 \Omega$ – stays constant
- Real part of the antenna: 7 Ω , 20 Ω , 30 Ω and 50 Ω
- Imaginary part of the antenna (from 160 Ω to 270 Ω)

Figure 3-9. Phase Angle $\Delta\phi$ versus Imaginary Part of the Antenna - Consideration only Valid for TSSOP10 Package

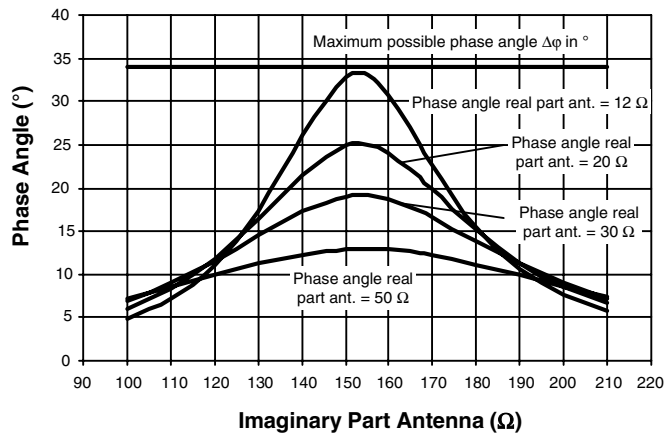


Figure 3-9 shows the phase angle $\Delta\phi$ (for the TSSOP10 package) versus the imaginary part of the antenna with the following conditions:

- Real part of the IC (state 1 and 2): $\approx 12 \Omega$ – stays constant
- Imaginary part of the IC (state 1): $\approx 159 \Omega$ – stays constant
- Imaginary part of the IC (state 2): $\approx 145 \Omega$ – stays constant
- Real part of the antenna: 12 Ω , 20 Ω , 30 Ω and 50 Ω
- Imaginary part of the antenna (from 100 Ω to 210 Ω)



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